DA3452 Speech and Language Technology

Module 6 Test

Preference learning, policy gradients, PPO and DPO

Duration 60 minutes Maximum marks 60 Questions 31

Name:	Roll No.:	Date:

Instructions

- Section A: 15 MCQs, 25 marks. Section B: 7 MSQs, 14 marks. Section C: 9 fill-in-the-blank questions, 21 marks. All questions are compulsory. Marks are shown against each question.
- MCQ: Select 1 option.
- MSQ: Select all correct options. An MSQ may have 1 or more correct options.
- MSQ marking: Selecting any incorrect option gives 0 for that question. Otherwise, marks are awarded in proportion to the correct options selected (rounded to the nearest 0.5). There is no negative marking.
- Record final answers on the response sheet. Each fill-in blank carries 1 mark.
- Use natural logarithms. The formula reference is at the end of this paper.
- **As announced, malpractice can result in disciplinary actions as per the Senate mandate.**

SECTION A MCQ

Questions 1-15 | 25 marks

1. **[1 mark]** Annotators compare responses R1 and R2. They prefer R1 because it is clearer. A later factual check finds an error in R1. What does the preference label tell us?

A. R1 was confirmed to be factually correct.

B. R1 was preferred to R2 in this pair.

C. R1 was preferred to responses the annotators did not see.

D. R1 gave the correct target token at each step.

2. **[1 mark]** An annotator sees 4 responses to the same prompt: R1, R2, R3 and R4. The annotator selects R3 as best. Which comparison is still unknown?

A. R3 versus R1.

B. R3 versus R4.

C. R1 versus R2.

D. R3 versus R2.

3. **[1 mark]** A service uses a fixed language model to generate 8 responses. It returns the response with the highest reward-model score. It computes no gradients and updates no parameters. What does the reward model do here?

A. It selects from the 8 generated responses.

B. It changes the generator's token probabilities.

C. It learns new scores from the 8 responses.

D. It gives the target token at each position.

4. [**2 marks**] For 1 fixed prompt, many judgements give these reliable preference probabilities: $P(R1 \succ R2)=0.7$, $P(R2 \succ R3)=0.7$ and $P(R3 \succ R1)=0.7$.

Can 1 set of scalar Bradley–Terry scores give all 3 probabilities exactly?

A. Yes. Add the same constant to all scores.

B. Yes. Set the 3 score gaps independently.

C. No. A probability of 0.7 requires a score of 0.7.

D. No. The 3 required score orderings contradict each other.

5. [**1 mark**] A frozen reward model gives fixed scores to 2 completed responses. A change to the generator's parameters changes their generation probabilities. Which quantities can change the expected reward?

A. The target scores used to train the reward model.

B. The stored preference labels from annotators.

C. The probabilities assigned to the 2 responses.

D. The scores assigned to the 2 fixed responses.

6. [**2 marks**] A support assistant receives 2 types of request. Password-reset requests make up 80%; billing requests make up 20%. The expected rewards are 1 for password resets and 3 for billing.

A reviewer tests 50 requests of each type. What is the expected reward for the actual mix of requests received by the assistant?

A. 2.0

B. 2.6

C. 1.0

D. 1.4

7. [**2 marks**] A support assistant can generate 2 complete responses to a password-reset request.

Response	Probability	Reward
H: "Here is a reset link."	p	5
R: "Please contact support."	$1-p$	2

The model uses $p=\sigma(z)$. The current value of p is 0.5.

A REINFORCE update samples R and uses no baseline. Use gradient ascent with a small positive learning rate. How do z and the expected reward change?

A. Increase z and increase the expected reward.

B. Decrease z and decrease the expected reward.

C. Increase z and decrease the expected reward.

D. Decrease z and increase the expected reward.

8. [**2 marks**] An email assistant has 2 possible replies to a fixed prompt. Each trial samples 1 reply. That reply is used to estimate 1 component of the policy gradient. The policy stays fixed across trials.

The engineer compares 2 methods. The table gives the mean and variance of the estimate over possible samples.

Method	Mean	Variance
Use the reward without a baseline	0.5625	0.0117
Subtract the exact mean reward as a baseline	0.5625	0.4219

Which statement is correct?

A. Both results can be correct. The baseline can increase the variance.

B. The baseline must depend on the reply sampled in that trial.

C. The 2nd result is wrong. Equal means require equal variances.

D. The 2nd result is wrong. The mean-reward baseline gives variance 0.

9. [**2 marks**] A team trains a summariser with REINFORCE. For each sampled summary, it records the reward and the sum of the token log-probabilities. The reward model is frozen.

During backpropagation, the code treats both values as constants. The policy gets no gradient. Which change fixes this error?

A. Keep both values fixed. Subtract the batch's mean reward.

B. Keep the log-probabilities fixed. Differentiate the reward through the reward model.

C. Keep the reward fixed. Recompute log-probabilities through the trainable policy.

D. Keep both values fixed. Increase the policy's learning rate.

10. [**2 marks**] A summariser has selected the token "postponed". The 3 policies give this token the probabilities below. All probabilities use the same preceding tokens.

Policy	Probability of "postponed"
Frozen SFT reference policy	0.20
Policy that generated the rollout	0.50
Current policy	0.60

Find the PPO probability ratio and the sampled log-ratio used for KL regularisation. Select the option with both values.

Option	PPO probability ratio	Sampled log-ratio for KL regularisation
A	3	$\ln 1.2$
B	1.2	$\ln 1.2$
C	3	$\ln 3$
D	1.2	$\ln 3$

11. **[2 marks]** A team compares policies P and Q for an email assistant. Both use the same frozen reference policy. The table gives each policy's expected reward and KL divergence from that policy to the reference.

Policy	Expected reward	KL divergence
P	1.80	0.10
Q	2.10	0.30

Use $\beta = 2$. Find the policy with the larger KL-regularised objective. Select that policy and its objective value.

Option	Policy with larger objective	Objective value
A	Q	1.50
B	P	1.60
C	P	1.80
D	Q	2.10

12. **[2 marks]** A summariser can generate 2 responses. F is a faithful summary with reward 2. U is an unsupported summary with reward -2.

The rollout policy gives each response probability 0.5. It generates the batch F, F, U, U. The current policy gives F probability 0.75.

Find the batch's mean reward without importance weights. Also find its importance-weighted estimate of the current policy's expected reward. Select the option with both values.

Option	Mean without importance weights	Importance-weighted estimate
A	0	1
B	1	0
C	0	0.5
D	0.5	1

13. **[1 mark]** A chatbot generates a rollout batch for PPO. It then updates the generator. The stored rollout probabilities stay fixed during these updates.

The updates on this batch are complete. Which model should generate the next rollout batch?

A. The frozen SFT reference.

B. The frozen reward model.

C. The updated generator.

D. The learned baseline predictor.

14. [2 marks] A source sentence says: "The launch moved from Monday to Friday." An annotator compares summaries F and U. The annotator prefers the faithful summary F.

Summary	Reference probability	Current probability
F: "The launch was postponed to Friday."	0.8	0.6
U: "The launch was completed on Friday."	0.2	0.4

Only these 2 responses are possible. DPO uses $\beta = 1$.

Find the current policy's probability of generating F. Also find the DPO model's predicted probability that an annotator prefers F to U. Select the option with both values.

Option	Probability of generating F	DPO-predicted probability of F $\succ$ U
A	0.6	0.6
B	$\frac{3}{11}$	0.6
C	0.6	$\frac{3}{11}$
D	0.4	$\frac{8}{11}$

15. [2 marks] A translation assistant has 2 possible replies to 1 request. F is a fluent translation. G is a garbled translation. Every preference label in its offline DPO dataset prefers F to G.

The reference gives both replies a nonzero probability. Keep $\beta > 0$ fixed. The policy can represent any probability between 0 and 1.

What happens to this pair's DPO loss as the odds of F increase?

A. It reaches its minimum when the policy equals the reference.

B. It approaches 0 as the odds of F grow without bound.

C. It reaches 0 when both replies have the same probability.

D. It reaches its minimum at the optimum for any fixed reward gap.

SECTION B MSQ

Questions 16-22 | 14 marks

16. [2 marks] An annotator prefers response R1 to R2. Call R1 the winner and R2 the loser. A team trains a reward model with pairs labelled in this way. Which statements are correct?

A. The same reward model scores both the winner and the loser.

B. The winner's score must be 1 because the annotator preferred it.

C. The reward model can score a new response that has no stored preference label.

D. A preference label gives the correct token at each generation step.

17. [**2 marks**] A writing assistant gets a reward-model score only when EOS ends its response. All earlier rewards are 0.

For 1 prompt, it generates exactly 3 tokens: “ready”, “.” and “EOS”.

Which statements are correct?

A. After “ready” is sampled, the next state is the prompt followed by “ready”.

B. The 0 reward after “ready” proves that this token did not help the response.

C. The episode has 3 token selections (actions), including EOS.

D. The reward-model score is given when EOS ends the response.

18. [**2 marks**] A team adds 7 to every reward-model score for 1 support prompt. The generator stays fixed.

Which quantities must stay unchanged?

A. Every REINFORCE gradient estimate based on 1 sampled response, without a baseline.

B. The Bradley–Terry probability for each response pair.

C. The exact gradient of expected reward with respect to the generator’s parameters.

D. The variance of the REINFORCE gradient estimate based on 1 sampled response, without a baseline.

19. [**2 marks**] A team trains a response generator with REINFORCE and a learned prompt-only baseline. Which choices are correct?

A. Predict the baseline before fitting it to the collected batch.

B. Hold the sampled advantages constant during the policy update.

C. Fit the baseline to the sampled response rewards.

D. Use each sample’s own response reward as its baseline.

20. [**2 marks**] Consider 1 PPO clipped-surrogate term. Use $\varepsilon = 0.2$. Keep the nonzero advantage $\widehat{A}$ fixed. Which statements are correct?

A. If $\widehat{A} > 0$ and $\rho = 0.6$, the term equals $0.6\,\widehat{A}$.

B. If $\widehat{A} < 0$ and $\rho = 1.4$, the term equals $1.4\,\widehat{A}$.

C. Every term is flat whenever ρ is outside [0.8, 1.2].

D. Optimising this objective forces every ratio to stay inside [0.8, 1.2].

21. [**2 marks**] Standard offline DPO uses a fixed set of preference pairs. Which statements are correct?

A. The frozen reference’s log-probabilities can be computed once for all pairs.

B. The current policy’s log-probabilities can be reused unchanged after every parameter update.

C. Computing the training loss needs no new responses from the current policy.

D. DPO trains the policy without fitting a separate reward model.

22. [**2 marks**] A team trains a chatbot with KL-regularised RLHF. The reward model and reference policy stay frozen.

Which statements are correct?

A. A sampled log-ratio $\ln\left[\dfrac{\pi_\theta(o \mid x)}{\pi_{ref}(o \mid x)}\right]$ can be negative.

B. Subtracting a prompt-only baseline removes the KL penalty from the expected objective.

C. The sampled adjusted reward is held constant during the score-function policy update.

D. Increasing β fixes incorrect reward-model rankings.

SECTION C FILL IN THE BLANKS

Questions 23-31 | 21 marks

23. **[2 marks]** 2 annotators see the same 6 responses to a prompt.

Annotator 1 ranks all 6 responses, with no ties. Annotator 2 selects only the best response.

How many winner–loser pairs can each annotation add to the reward-model dataset?

(a) Pairs from Annotator 1's full ranking: _____

(b) Pairs from Annotator 2's best-only choice: _____

24. **[2 marks]** 6 annotators compare a clear reply C with a vague reply V. 5 prefer C; 1 prefers V.

A Bradley–Terry model fits only this pair. The score gap $\delta = r(C) - r(V)$ can take any real value.

Find the preference probability and score gap that minimise the average comparison loss. You may use exact fractions and logarithms.

(a) $P(C \succ V)$: _____

(b) Score gap δ: _____

25. **[2 marks]** A completed response has 2 tokens: "yes", then "EOS". The probability of "yes" given the prompt is 0.4.

Before an update, $P(EOS \mid prompt, yes) = 0.5$. The update changes only this conditional probability to 0.75.

Find the completed-response probability before and after the update.

(a) Before: _____

(b) After: _____

26. **[2 marks]** A summariser can generate 2 responses. F is a faithful summary with reward 3. U is an unsupported summary with reward 1.

The probability of F is $p = \sigma(z)$. The current value of p is 0.5. There are 3 independent samples: F, F, U.

Use REINFORCE without a baseline. Find the exact expected-reward gradient with respect to z. Also find the gradient estimate from this batch.

(a) Exact expected-reward gradient $\frac{dJ}{dz}$: _____

(b) Gradient estimate from the batch: _____

27. **[2 marks]** A team estimates 1 component of the policy gradient. It samples responses independently and averages their gradient estimates. The policy stays fixed. Each estimate from 1 response has variance 0.64.

(a) Variance of the mean for a batch of 16 samples: _____

(b) Total batch size needed to halve the standard deviation of the mean, compared with the batch of 16:

_____

28. [**2 marks**] A summariser can generate 2 responses: a faithful summary F or an unsupported summary U.

Summary	Reference probability	Current probability
F	0.5	0.8
U	0.5	0.2

The reward-model score for U is 1. Use $\beta = 2$.

Use $\ln 1.6 = 0.4700$ and $\ln 0.4 = -0.9163$. Give answers to 4 decimal places.

(a) KL divergence $D_{KL}(\pi_\theta \,\|\, \pi_{ref})$, in nats: _____

(b) Adjusted reward for a sampled U: _____

29. [**2 marks**] For 1 stored PPO sample, use these fixed values:
$\widehat{A} = -2$, $\rho = 0.6$ and $\varepsilon = 0.2$.

Here, ρ is the current probability divided by the rollout probability. Find the unclipped product. Then apply the minimum in the PPO formula to find the clipped-surrogate term.

(a) Unclipped product: _____

(b) Final clipped-surrogate term: _____

30. [**2 marks**] An email assistant has 2 possible complete responses: helpful H and unhelpful U. The reference gives H probability 0.2 and U probability 0.8. The fixed reward gap is $r(H) - r(U) = \ln 4$.

Find the optimal probability of H for the KL-regularised objective at each value of β below.

You may use fractions.

(a) $\beta = 1$: _____

(b) $\beta = 2$: _____

31. [**5 marks**] A support model has 2 possible responses. Response E passes the request to a staff member. Response D gives a direct answer.

The probability of E is $p = \sigma(z)$. The probability of D is $1 - p$. E has fixed reward u. D has fixed reward v.

Complete parts (a)–(c) for the expected reward. In parts (d)–(e), the model samples D for 1 REINFORCE update. Use a prompt-only baseline b and a positive learning rate α. Hold u, v and b fixed. Use the value of p before the update.

Write expressions using p, u, v, b, z and alpha (or α). Use p in your answers instead of $\sigma(z)$.

(a) Expected reward $J(p)$ = _____

(b) Derivative $\dfrac{dJ}{dp}$ = _____

(c) Use the chain rule: $\dfrac{dJ}{dz}$ = _____

(d) Sampled REINFORCE gradient estimate for D (baseline b), g = _____

(e) New logit after 1 gradient-ascent update, z_{new} = _____

FORMULA REFERENCE

All logarithms are natural. A completed response o includes EOS. In the expressions below, x is a prompt, π_θ is the trainable policy, and $r(x,o)$ is a fixed reward during a policy update. The learning rate is $\alpha>0$. The reference-penalty coefficient is $\beta>0$.

1. Sigmoid and elementary derivatives

$$p=\sigma(z)=\frac{1}{1+\exp(-z)},\frac{dp}{dz}=p(1-p).$$

$$\frac{d\ln p}{dz}=1-p,\frac{d\ln(1-p)}{dz}=-p.$$

If $q=\sigma(\delta)$, then $\delta=\ln\left[q/(1-q)\right]$. Also $\sigma(\ln a)=a/(1+a)$ for $a>0$.

2. Preference comparisons

For N responses with no ties: a full ranking gives $N(N-1)/2$ pairs; 1 best-only choice gives $N-1$ pairs.

$$\delta=r(x,o_w)-r(x,o_l),P(o_w\succ o_l\mid x)=\sigma(\delta).$$

$$L_{pair}=-\ln\sigma(\delta),\frac{dL_{pair}}{d\delta}=\sigma(\delta)-1.$$

For n_A labels preferring A and n_B labels preferring B, with $q=P(A\succ B\mid x)$:

$$L(q)=-n_A\ln q-n_B\ln(1-q).$$

3. Response probability and expected reward

$$\pi_\theta(o\mid x)=\prod_{t=1}^{T}\pi_\theta(o_t\mid x,o_{<t}).$$

$$\ln\pi_\theta(o\mid x)=\sum_{t=1}^{T}\ln\pi_\theta(o_t\mid x,o_{<t}).$$

$$J_x(\theta)=\sum_o\pi_\theta(o\mid x)r(x,o),J(\theta)=E_{x\sim D}\left[J_x(\theta)\right].$$

4. REINFORCE, baselines and updates

$$\nabla_\theta\pi_\theta(o\mid x)=\pi_\theta(o\mid x)\nabla_\theta\ln\pi_\theta(o\mid x).$$

For independent responses o_i sampled from the current policy:

$$\hat{g}=\frac{1}{N}\sum_{i=1}^{N}\left[r(x_i,o_i)-b(x_i)\right]\nabla_\theta\ln\pi_\theta(o_i\mid x_i).$$

Gradient ascent: $\theta_{new}=\theta+\alpha\hat{g}$.

The baseline $b(x)$ depends only on the prompt. During the policy update, hold the sampled reward and baseline fixed. Fit a learned baseline with

$$L_b=\frac{1}{N}\sum_{i=1}^{N}\left[b_\psi(x_i)-r(x_i,o_i)\right]^2.$$

5. Variance of a sample mean

Consider N independent scalar estimates with the same distribution. Each estimate has variance V:

$$Var(\text{mean})=\frac{V}{N},SD(\text{mean})=\sqrt{\frac{V}{N}}.$$

FORMULA REFERENCE CONTINUED

6. Reference policy and KL regularisation

$$\ell_\theta(x,o)=\ln\left[\frac{\pi_\theta(o \mid x)}{\pi_{ref}(o \mid x)}\right].$$

$$D_{KL}(\pi_\theta \,\|\, \pi_{ref})=\sum_o \pi_\theta(o \mid x)\,\ell_\theta(x,o).$$

$$R_\theta(x,o)=r(x,o)-\beta\,\ell_\theta(x,o).$$

$$J_\beta(\theta)=E_{x\sim D}\left[E_{o\sim\pi_\theta}\left[r(x,o)\right]-\beta\,D_{KL}(\pi_\theta \,\|\, \pi_{ref})\right].$$

$$J_\beta(\theta)=E_{x,o}\left[R_\theta(x,o)\right].$$

7. Importance weighting and PPO

Sample completed responses from π_{old}. Estimate an expectation under the current policy with

$$\frac{1}{N}\sum_{i=1}^{N}\frac{\pi_\theta(o_i \mid x)}{\pi_{old}(o_i \mid x)}f(o_i).$$

For PPO, $s_t=(x,o_{<t})$ and the sampled token is o_t:

$$\rho_t=\frac{\pi_\theta(o_t \mid s_t)}{\pi_{old}(o_t \mid s_t)}.$$

$$\text{Unclipped term}=\rho_t\widehat{A}_t.$$

$$\text{Clipped term}=min\left[\rho_t\widehat{A}_t, clip(\rho_t,1-\varepsilon,1+\varepsilon)\widehat{A}_t\right].$$

$$clip(\rho,l,u)=min(max(\rho,l),u).$$

PPO maximises the average clipped surrogate. Keep the stored advantages fixed during these updates.

8. KL-regularised optimal policy

For a fixed reward and reference:

$$\pi^*(o \mid x)=\frac{\pi_{ref}(o \mid x)\exp\left[r(x,o)/\beta\right]}{Z(x)},$$

where

$$Z(x)=\sum_o \pi_{ref}(o \mid x)\exp\left[r(x,o)/\beta\right].$$

Thus, for any 2 responses A and B:

$$\frac{\pi^*(A \mid x)}{\pi^*(B \mid x)}=\frac{\pi_{ref}(A \mid x)}{\pi_{ref}(B \mid x)}\exp\left[\frac{r(x,A)-r(x,B)}{\beta}\right].$$

9. DPO

For a recorded preferred response o_w and rejected response o_l:

$$m_\theta=\beta\left\{\ln\left[\frac{\pi_\theta(o_w \mid x)}{\pi_{ref}(o_w \mid x)}\right]-\ln\left[\frac{\pi_\theta(o_l \mid x)}{\pi_{ref}(o_l \mid x)}\right]\right\}.$$

$$\text{Predicted preference probability}=\sigma(m_\theta), L_{DPO}=-\ln\sigma(m_\theta).$$

Response log-probabilities sum over the response tokens, including EOS. Prompt and padding tokens are excluded.